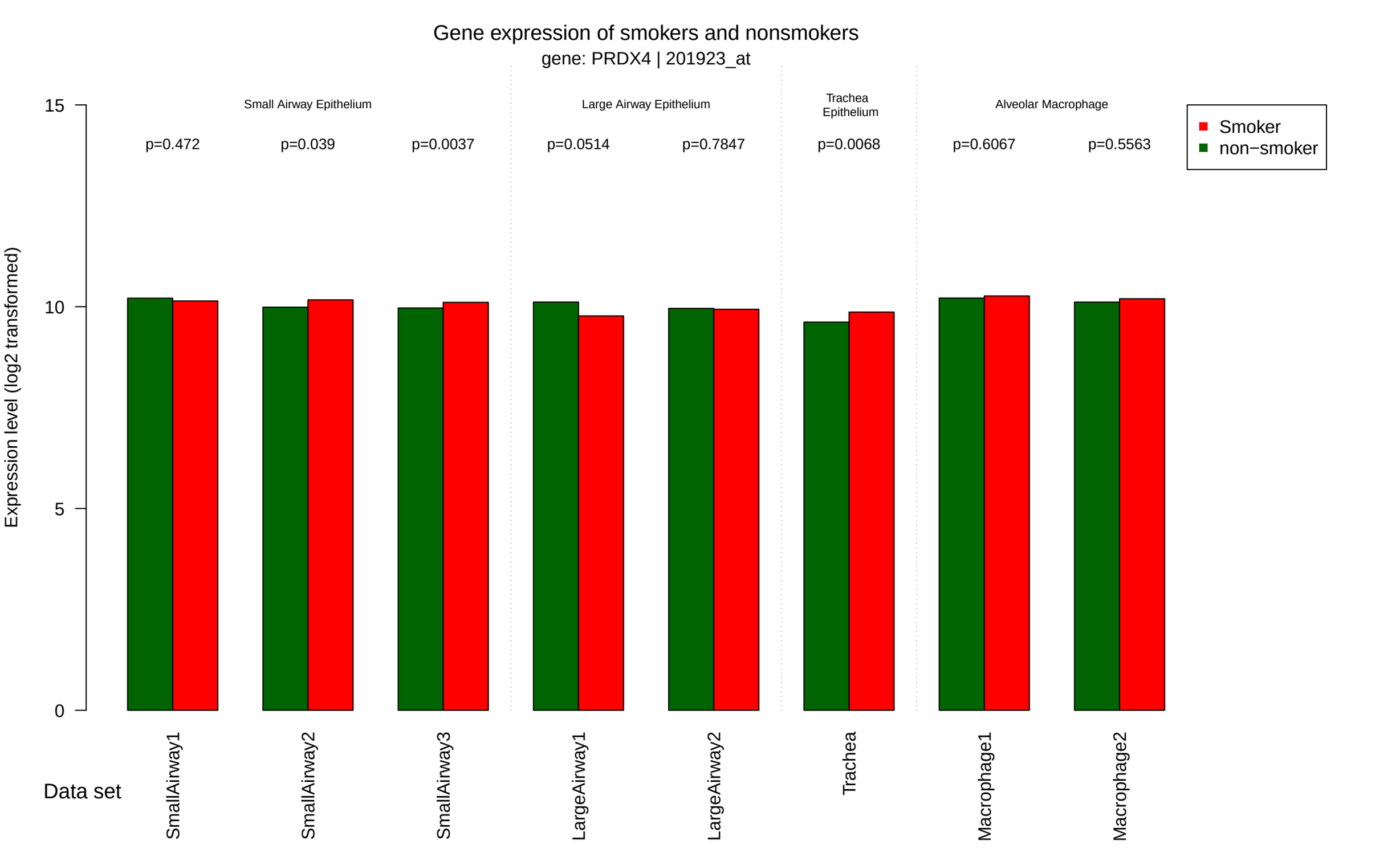
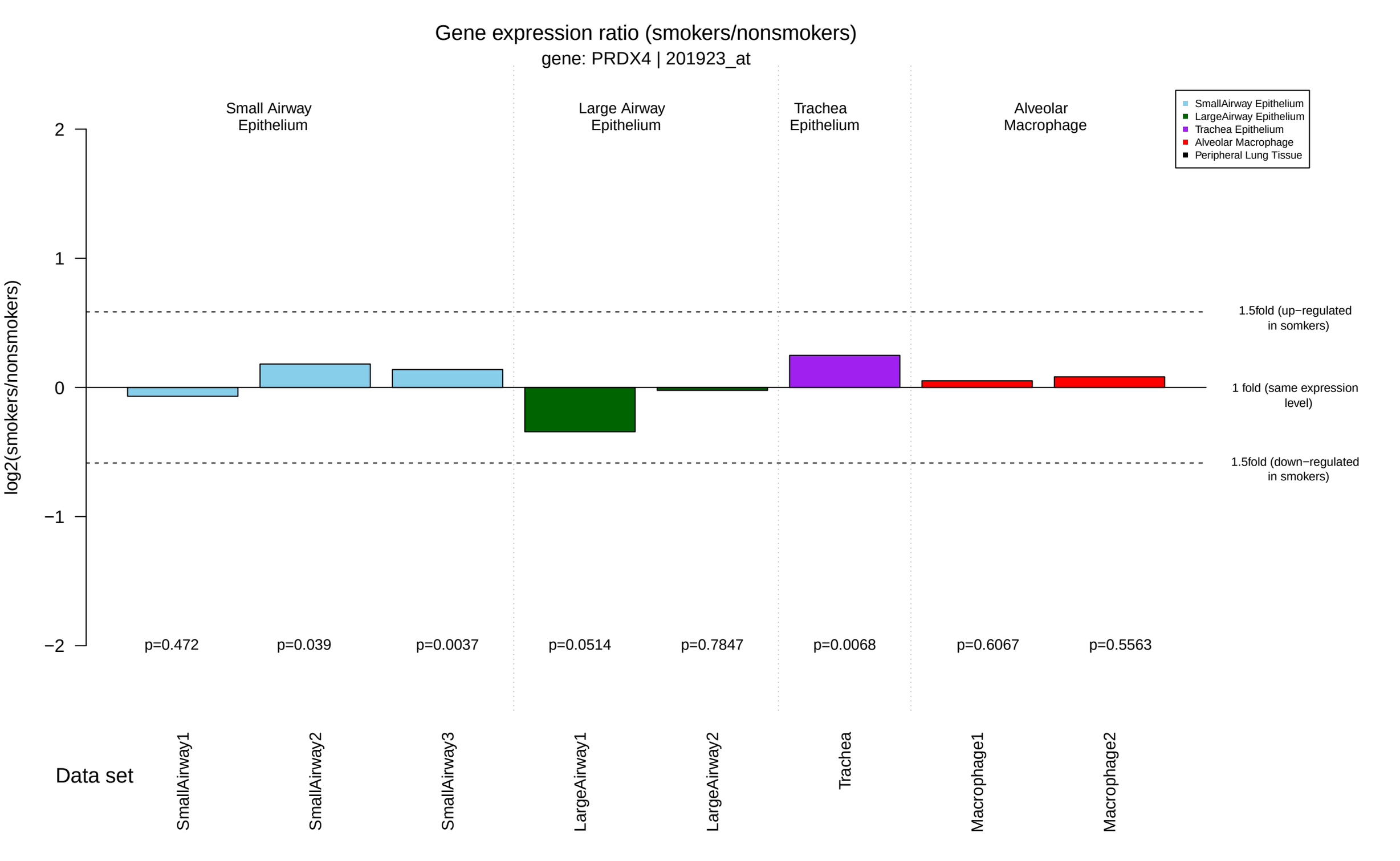
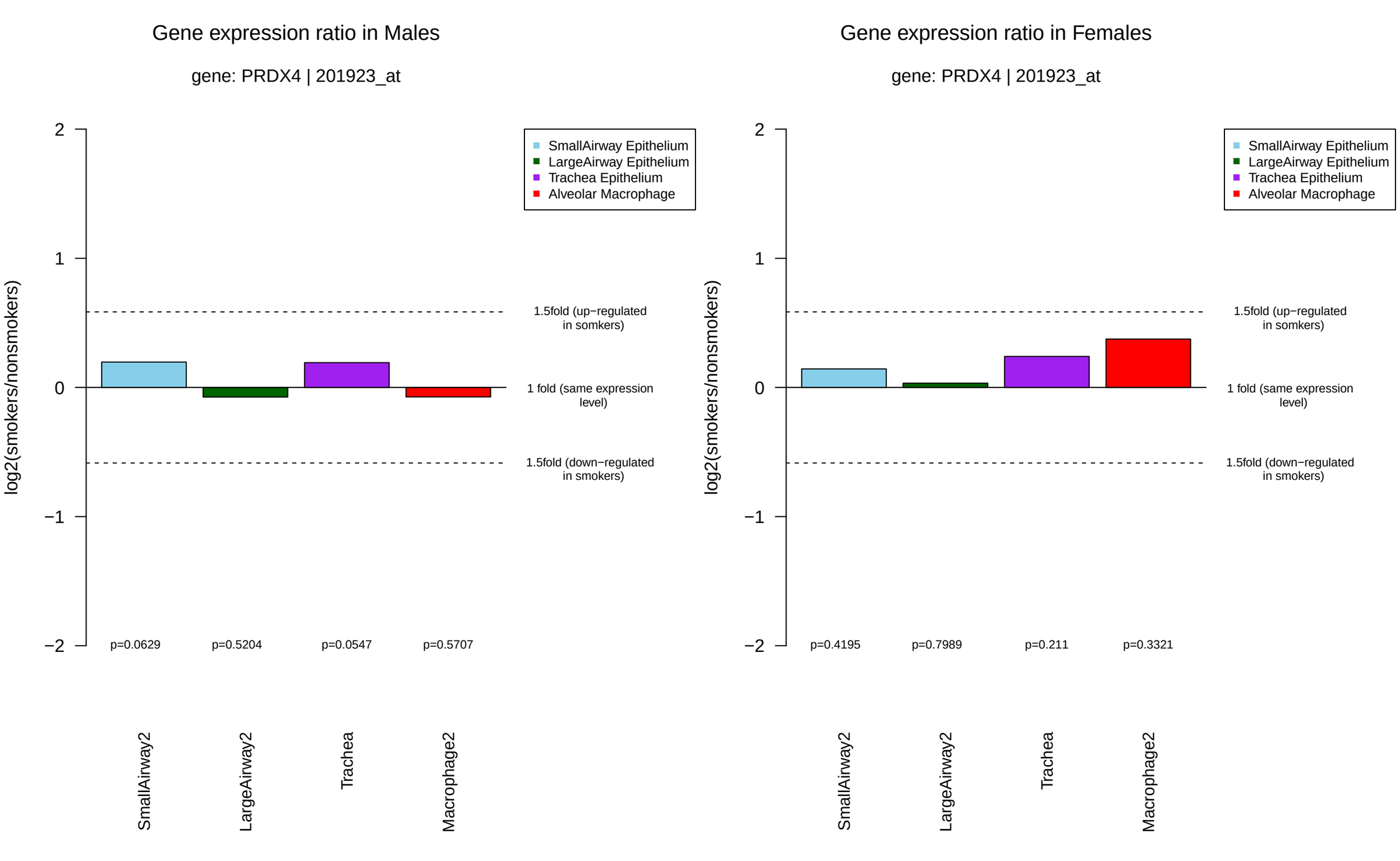
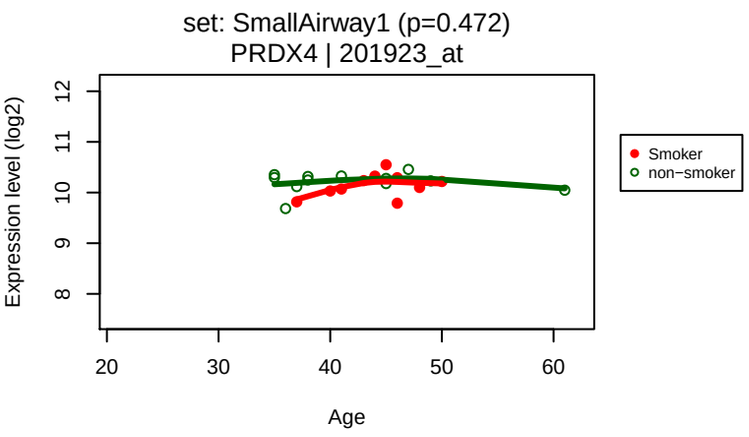
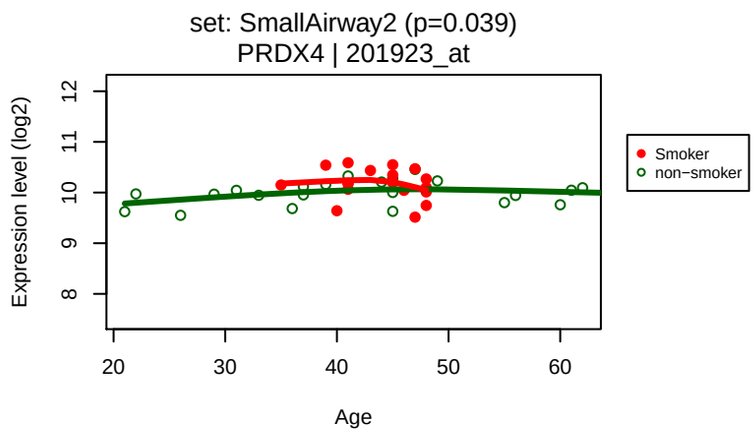
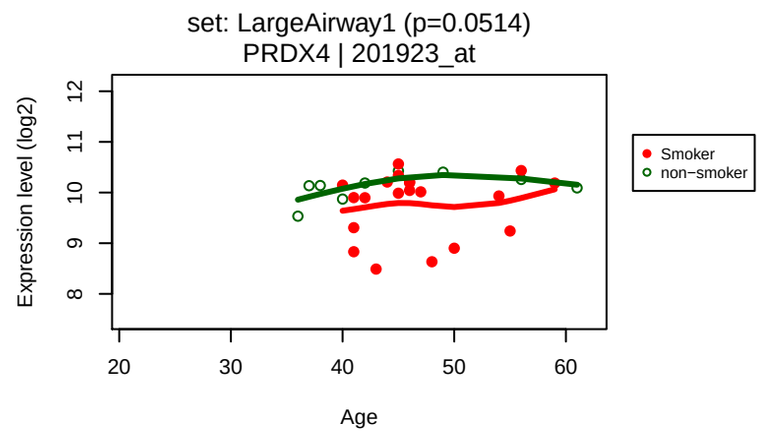

Gene expression of smokers and nonsmokers
gene: PRDX4 | 201923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.472
p=0.039
p=0.0037
p=0.0514
p=0.7847
p=0.0068
p=0.6067
p=0.5563
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

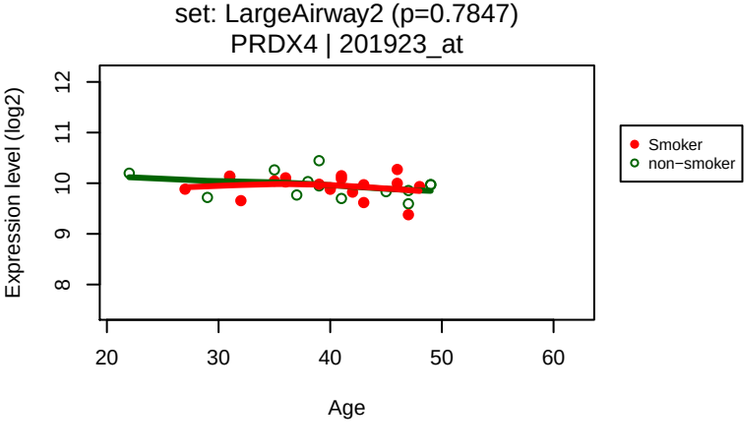

Gene expression ratio (smokers/nonsmokers)
gene: PRDX4 | 201923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.472
p=0.039
p=0.0037
p=0.0514
p=0.7847
p=0.0068
p=0.6067
p=0.5563
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

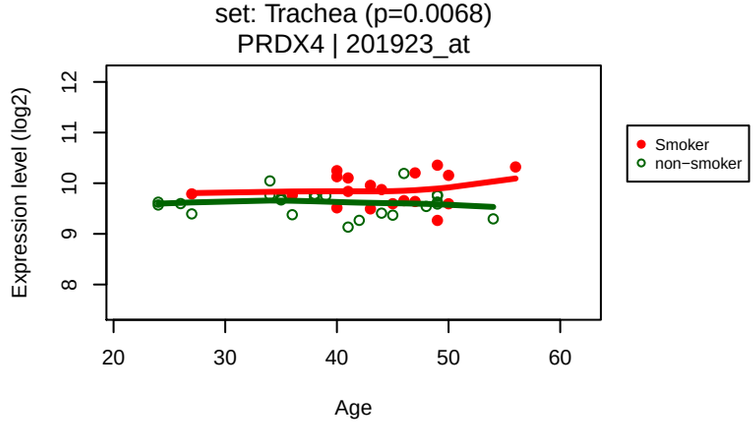

Gene expression ratio in Males
gene: PRDX4 | 201923_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0629
p=0.5204
p=0.0547
p=0.5707
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRDX4 | 201923_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4195
p=0.7989
p=0.211
p=0.3321
SmallAirway2
LargeAirway2
Trachea
Macrophage2

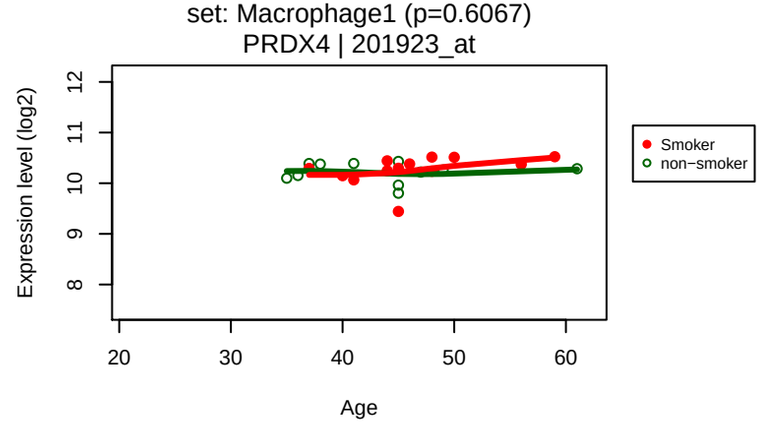

set: SmallAirway1 (p=0.472)
PRDX4 | 201923_at
Expression level (log2)
Age
Smoker
non−smoker

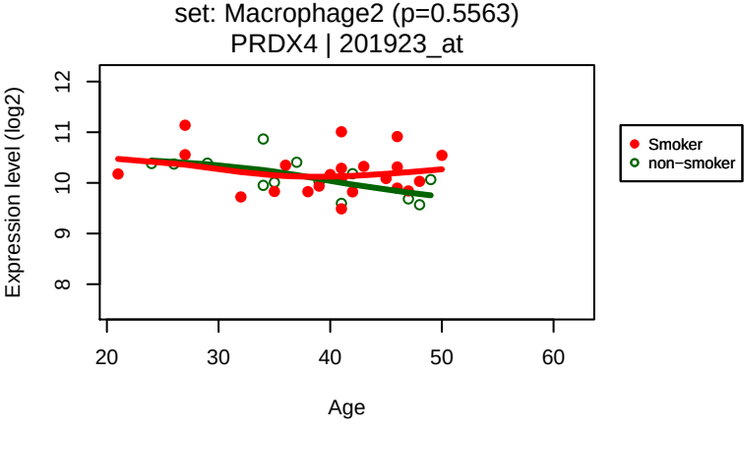

set: SmallAirway2 (p=0.039)
PRDX4 | 201923_at
Expression level (log2)
Age
Smoker
non−smoker

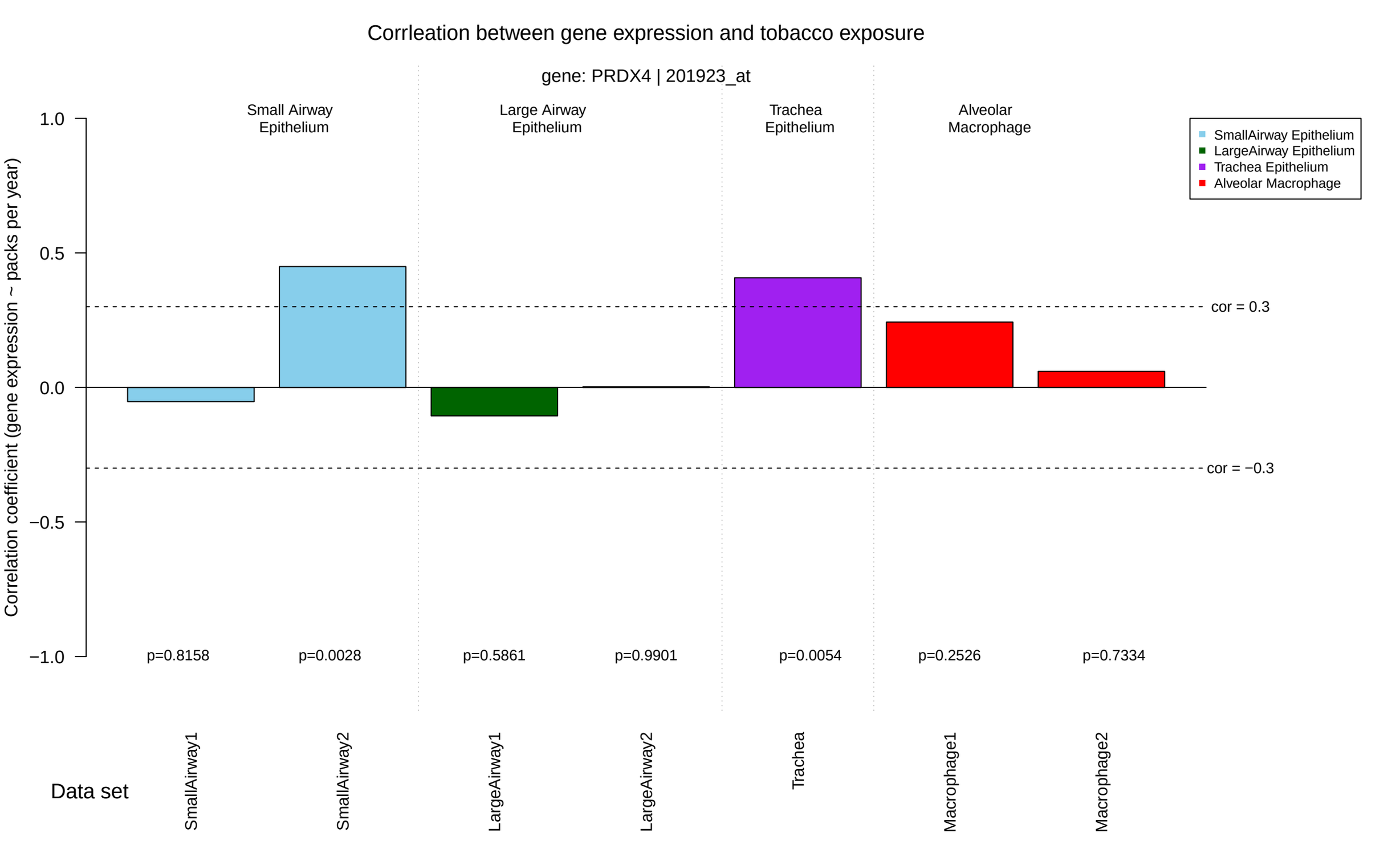

set: LargeAirway1 (p=0.0514)
PRDX4 | 201923_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7847)
PRDX4 | 201923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0068)
PRDX4 | 201923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6067)
PRDX4 | 201923_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5563)
PRDX4 | 201923_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRDX4 | 201923_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8158
p=0.0028
p=0.5861
p=0.9901
p=0.0054
p=0.2526
p=0.7334
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2